

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration: 1 Hour

Total Marks: 50

Annexure A

Scenario Based Assignment

Code of Conduct:

I R. Rupa Sree (1980470182) (Name/ Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Rupa Sree

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B. N. J.

1. Scenario

An airline implements an AI system to optimize crew scheduling across hundreds of flights and thousands of crew members. The objective is to minimize crew costs, layover time, and scheduling conflicts while satisfying regulatory rest requirements. The system must continuously adjust assignments as flight delays occur. However, the search space is extremely large, flight schedules change dynamically, and the system may get stuck in locally optimal crew assignments.

Tasks: Formulate this as an optimization problem and explain why traditional search is inefficient.

- i. Explain how Hill Climbing would work in this scenario and identify its limitations.
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation.
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations.
- iv. Recommend the best approach and justify your choice based on scalability and adaptability.

2. Scenario

A firefighting robot is deployed inside a burning multi-store building with an unknown internal layout. It must: navigate safely to locate trapped occupants, avoid hazards (collapsed structures, fire zones), operate with limited visibility due to smoke, and make decisions without a complete floor plan. The robot updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search.
- ii. Analyse the challenges of partial observability and dynamic environments.
- iii. Describe how the robot builds and updates its internal model.
- iv. Compare online search with uninformed and informed search strategies.
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety.

3. Scenario

An airport is designing an AI system to automatically assign arriving flights to available gates. The system must satisfy multiple constraints: no two flights can occupy the same gate at overlapping times, a limited number of gates support wide-body aircraft, certain airlines have contractual gate preferences, ground crew availability constraints, and special accommodations for delayed connecting flights. The complexity increases as the number of flights and gates grows.

Tasks: Formulate the problem as a Constraint Satisfaction Problem (CSP).

- i. Clearly define variables, domains, and constraints with explanation.
- ii. Explain how backtracking search works in this scenario.
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency.
- iv. Analyse real-world challenges and justify the best strategy for scalability.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

$$20 + 20 + 25 + 20 + 15 = 80$$

1. Airline Crew Scheduling - Optimization

Problem Formulation:-

The airline crew scheduling problem can be formulated as optimization problem.

Objective Function:-

Minimize:

- To crew salary and overtime costs
- Layover distribution
- Scheduling conflicts
- Flight delays caused by crew unavailability

Subject to:

- Rest Requirements
- Maximize working hours
- Crew availability and qualifications
- All flights being covered.

Traditional Search is inefficient because there are thousands of crew members and flights, creating a huge number of possible schedules.

1) Hill Climbing:-

Hill climbing starts with a schedule and repeatedly makes small change to improve the solution.

Steps:

1. Create a initial schedule.
2. Change a crew assignment.
3. If the new schedule is better, accept it.
4. Continue until no improvement is possible.

Limitations:

- Can stop before reaching the best schedule.
- Cannot recover after choosing a poor direction.
- Easily trapped in local optimum.

ii) Local Maxima, Plateaus and Ridges:-

1) Local Maximum:-

The algorithm reaches a schedule where every nearby change increase cost. Although a better schedule exists elsewhere, Hill climbing cannot reach it.

2) Plateau:-

Many neighboring schedule have identical cost. The algorithm has no direction for improvement.

3) Ridge:-

The optimal direction requires several coordinated changes. Single crew swaps do not improve the schedule.

iii) Simulated Annealing and Genetic Algorithm:-

Simulated Annealing:-

works like Hill climbing but occasionally accept worse solution.

Advantages:-

- Escapes local Maxima
- Better exploration
- Suitable for dynamic optimization

Eg:- Accept slightly worse schedule temporarily to discover a much better schedule later.

Genetic Algorithm:-

Inspired by natural evolution

Steps:

- 1) Generate many schedule
- 2) Evaluate fitness.
- 3) Select best schedule
- 4) perform crossover
- 5) Apply mutation
- 6) Repeat until convergence

Advantages:

- Highly scalable
- Avoids premature convergence
- Works well for very large search space

iv) Best Recommendation:-

Genetic Algorithm:-

It is recommended because it:

-) Handles very large search space
-) Avoids local optima.
-) Support multiple constraints
-) Can adapt when flight schedule change.

2. Firefighting Robot (online Search Agent)

1) Why Online search instead of offline search?

Offline search assumes:

-) complete map search
-) Environment does not change.

Firefighting robot has:

-) Unknown building layout
-) Smoke reducing visibility

Therefore it cannot plan either path beforehand.

Instead it must:

-) Observe •) Learn
-) Move •) Replan continuously

Hence an Online Search Agent is required.

2) Challenges:-

Partial Observability:

-) Robot cannot see the entire building.
-) Smoke hides obstacles and people.

Dynamic Environment:

-) Fire spreads
-) Structure may collapse.
-) Safe path become dangerous

3) Internal Model :-

The robot uses sensors to collect information and builds a map of :

-) Visited areas
-) Obstacles
-) Free zone
-) Safe paths
-) Victim location

It continuously updates this map as a new information is received.

4) Comparison :-

Feature	Online Search	Uninformed Search	Informed Search
Unknown environment	Yes	No	Limited
Learns while moving	Yes	No	No
Adapts to changes	High	Low	Moderate
Suitable for robots	Yes	No	Limited.

5) Best Approach :-

Online Search Agent :-

Advantages :-

-) Learns continuously
-) Adapts to new hazards
-) Make decision using current observations
-) Handles uncertainty effectively
-) Provides safer navigation

This makes it ideal for firefighting robots.

8. Airport Gate Assignment (CSP) :-

Problem Foundation

The airport gate assignment problem is modelled as constraint satisfaction problem.

Goal :- Assign every arriving flight to suitable gate while satisfying all constraints.

i) Variables, Domains and Constraints :-

Variables :-

Each arriving flight is a variable.

eg:- $F_1, F_2, F_3, \dots, F_n$

Domains :-

Possible gates for each flight.

eg:- $F_1 \rightarrow \{G_1, G_2, G_3\}$

$F_2 \rightarrow \{G_2, G_4\}$

$F_3 \rightarrow \{G_1, G_5\}$

Constraints :-

- Two flights cannot use the same gate at the same time.
- Wide body aircraft need suitable gates.
- Airline gate preference should be satisfied.
- Ground crew must be available.
- Connecting flights may require nearby gates.

ii) Backtracking Search :-

Backtracking assigns gates one flight at a time.

eg:- Assign $F_1 \rightarrow G_1 \checkmark$

$F_2 \rightarrow G_2 \checkmark$

$F_3 \rightarrow G_2 \times$

conflict occurs

Back track: Assign

$F_3 \rightarrow G_4 \checkmark$

continue until every flight receives valid gate

000 Constraint Propagation (Forward checking)

↳ Forward checking removes invalid gate choice immediately.

Eg:- Suppose $F1 \rightarrow G1$

Time: 10:00-11:00

THEN: Remove G1 from all overlapping flights

This prevents future conflicts early

Advantages:-

- ↳ Reduce unnecessary search
- ↳ Improve efficiency.

IV Real-world challenges and Best Strategy

↳ challenges:-

- ↳ flight delays
- ↳ Gate maintenance
- ↳ Emergency Analysis
- ↳ weather disruptions
- ↳ Large number of flights
- ↳ Limited gate availability

Best Strategy:-

constraint satisfaction (CSP) with Backtracking and Forward checking (constraint propagation)

Advantages:

- ↳ Efficiently handles complex constraints
- ↳ Eliminates invalid assignment early.
- ↳ Reduce search time
- ↳ Scales better for large airports

This combination provides an effective balance between correctness, efficiency and scalability for real-world airport gate assignment systems.